Mostakim Khan Rayan

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GitHub | Portfolio | LinkedIn

FOUNDER WORK

Sleight AI — Personal AI Entity & Automated Builder
2026–Present

Creator and builder | In development

- **Building personal AI from a user's chosen data.** Built A.L.I.C.E. as the first development case with versioned memory, source-linked evidence, owner permissions and traceable correction/deletion.
- **Trained a 267.5M-parameter conditional decoder** over frozen Gemma features: 20 epochs on 36,597 public pairs. Reloaded a 278.4M successor; its new reference/copy heads remain in progress.
- **Trained a three-stage personality candidate** for evidence representation, judgment and calibration. Reproduced all 60 saved decisions after checkpoint reload; broader behavioral validation remains pending.
- **Built FloRA's durable comparison infrastructure.** Verified 1,047 contract tests and 38 core integration cases, including restart recovery. The full behavioral comparison remains in progress. FloRA

SELECTED RESEARCH AND ENGINEERING EXPERIENCE

AFM Morphology Analysis Platform
Nov. 2025–Present

AI Systems Integration Engineer | Dr. Boran Ma's Research Lab

- **Shipped a researcher-facing AFM platform.** Integrated classification, segmentation and morphology analysis through Flask/Next.js, mask correction and PDF reports. Cut analysis turnaround about 65%.
- **Released a 50-image benchmark** with 90.0% classification accuracy and 0.845 mean Dice. A measured batch ran at 6.66 images/minute. Published evaluation data and scripts. Dataset

Molecular Dynamics & Scientific Computing
2026–Present

Undergraduate Research Assistant | Dr. Sergei Nazarenko's Research Group

- **Built LAMMPS/HPC workflows for shock-wave polymer research.** Automated density profiles across six polyisobutylene blends, with Slurm batch jobs, restart handling and equilibration checks.

Machine Learning Intern — Vision AI Development
May–Aug. 2025

- **Deployed a financial-document vision pipeline.** Reached 93–96% accuracy on 15,000 held-out documents, reduced manual review about 50% and cut inference latency over 30%.

Open Source GPU Software
Aug. 2026–Present

External contributor | AMD ROCm Iris | NVIDIA RAFT | RAPIDS cuML

- **Merged GPU contributions into AMD ROCm Iris and NVIDIA RAFT.** Jacobi reference tests passed at 2, 4 and 8 ranks; the CUDA indexing fix includes float/double and memory-layout regression tests.

Three additional cuML PRs cover GPU data generation, IVF configuration and multiclass workflows;

EARLY BUILDING AND LEADERSHIP

- **Deployed an automated water-reservoir controller for my community.** Adapted my robotics competition prototype to stop tank overflow in a building housing 64 families.
- **Mentored 150+ robotics participants at Roboment (2021).** Led training and project troubleshooting.

PUBLICATIONS AND MANUSCRIPTS

Morphology-aware classification, segmentation, and downstream analysis of Block Copolymer AFM images — First and corresponding author. SoftwareX (under peer review); SSRN preprint available.

Two first-author, peer-reviewed high-school papers in the Journal of Emerging Investigators: fluorinated allopurinol (2022) and the Mpemba effect (2023).

EDUCATION AND SELECTED HONORS

Vanderbilt University | B.S. Electrical and Computer Engineering | Expected May 2028 | GPA 3.94

Team Gold, International Robotics Olympiad (2022–23) | Team Silver, International Earth Science Olympiad (National Field Team round)

Gold, Queen's Commonwealth Essay Competition | Gold, The Duke of Edinburgh's International Award

Exchange student: Asia Kakehashi, Japan; Kennedy-Lugar YES, Wisconsin, USA